EE671 VLSI Design (Course Project-1)
Team Number 23

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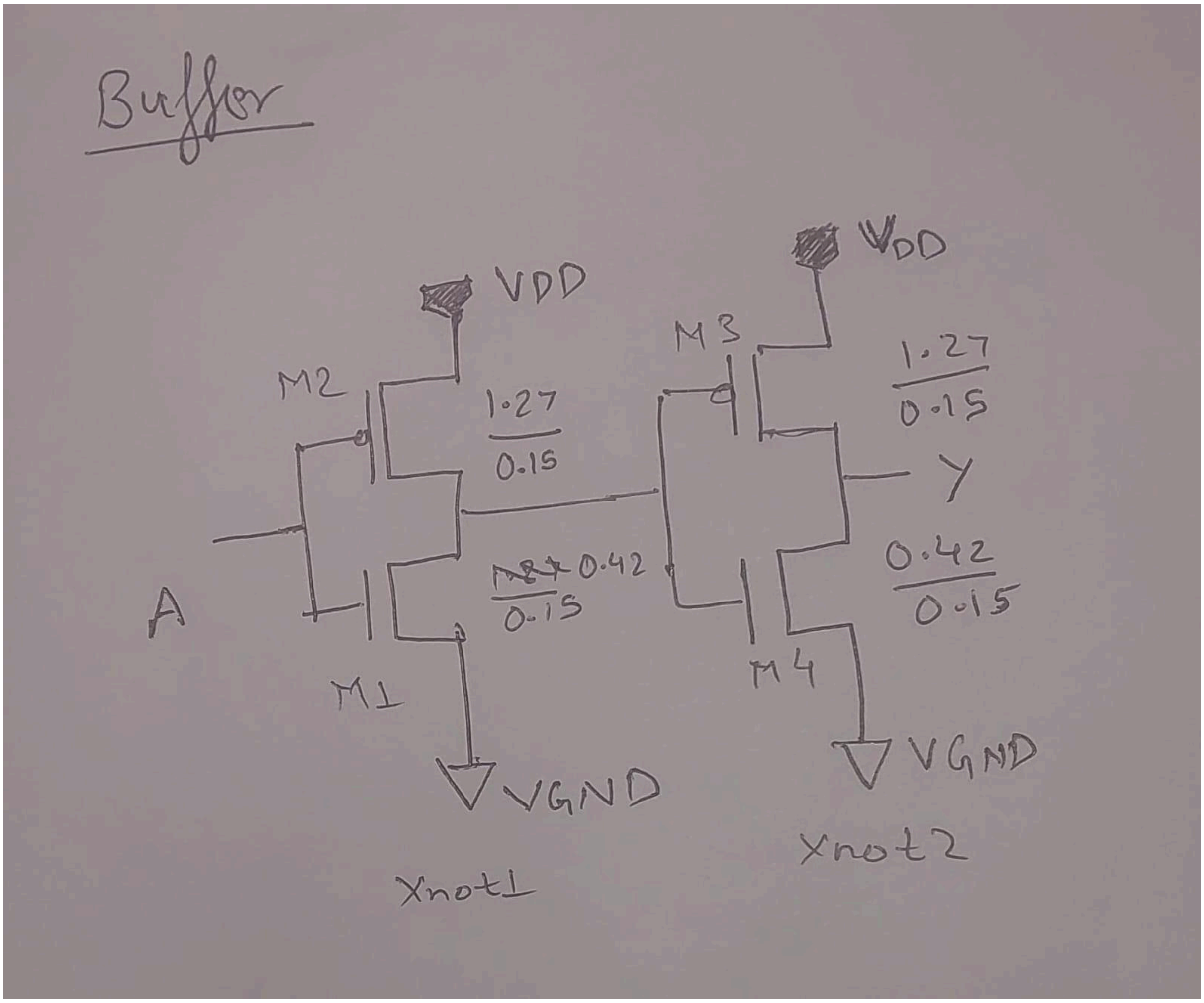
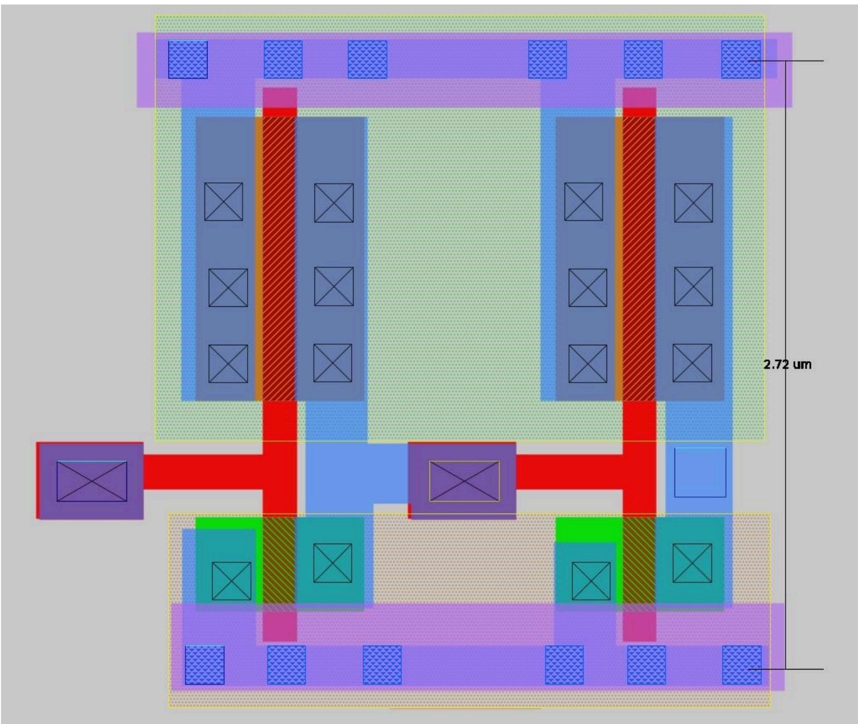
Buffer
Circuit Diagram

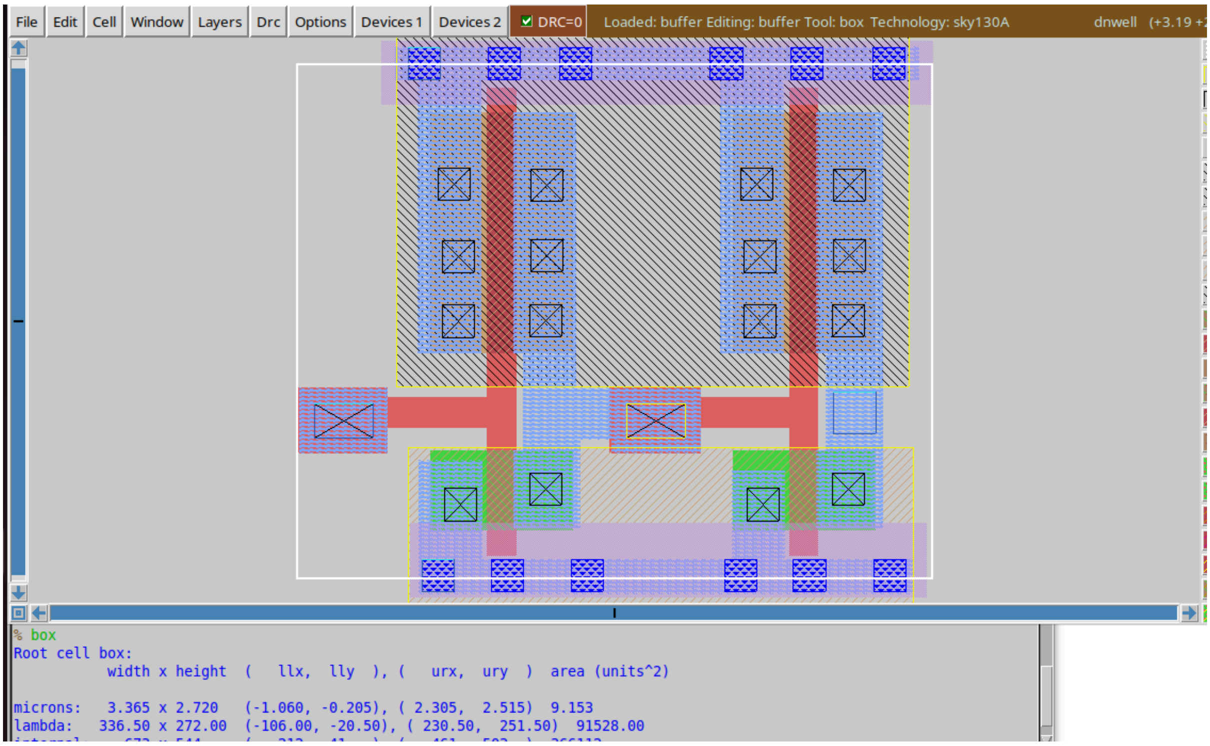


	W(in um)	L(in um)
M1	0.42	0.15
M2	1.27	0.15
M3	1.27	0.15

M4	0.42	0.15

Layout Schematic





Area of cell = width x height = 3.365 um x 2.272 um

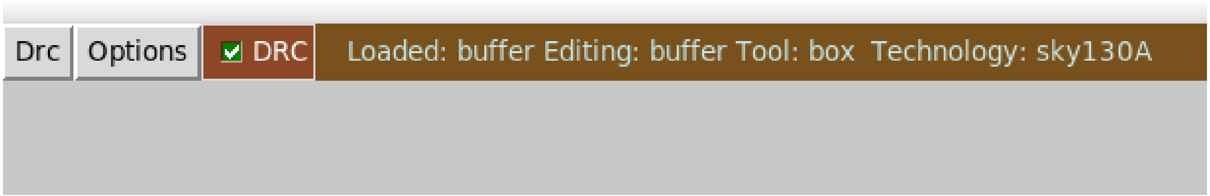
Pex netlist screenshot

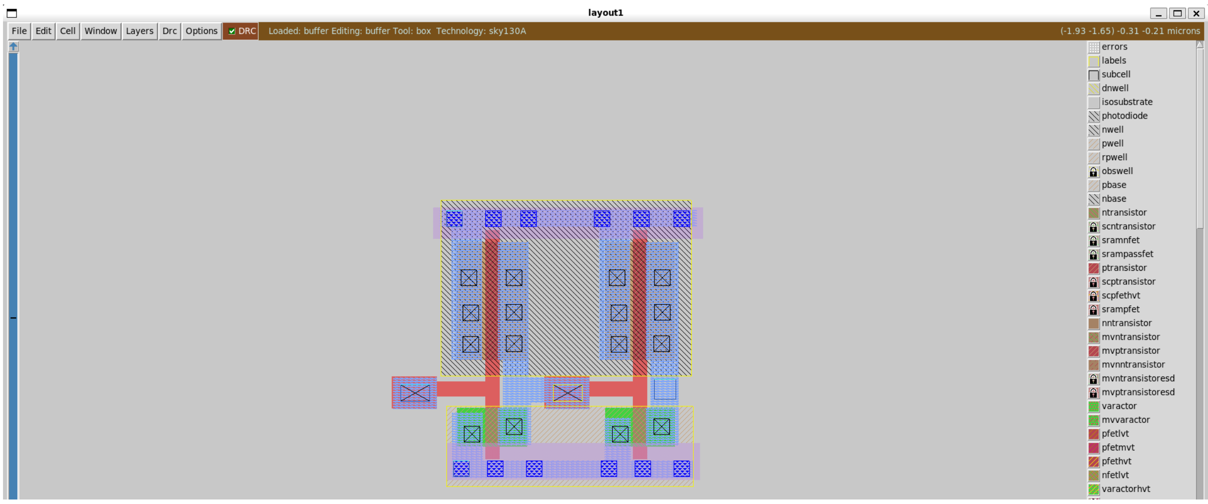
```
* NGSPICE file created from copy.ext - technology: sky130A

.subckt copy A VDD VGND Y
X0 out1 A VDD VDD sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X1 Y out1 VDD VDD sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X2 Y out1 VGND VGND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X3 out1 A VGND VGND sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
C0 VDD Y 0.1642f
C1 VDD out1 0.33741f
C2 VDD A 0.07945f
C3 Y out1 0.04407f
C4 A out1 0.05503f
C5 Y VGND 0.17045f
C6 VDD VGND 1.07612f
C7 A VGND 0.31f
C8 out1 VGND 0.37927f
.ends
```

LVS Check

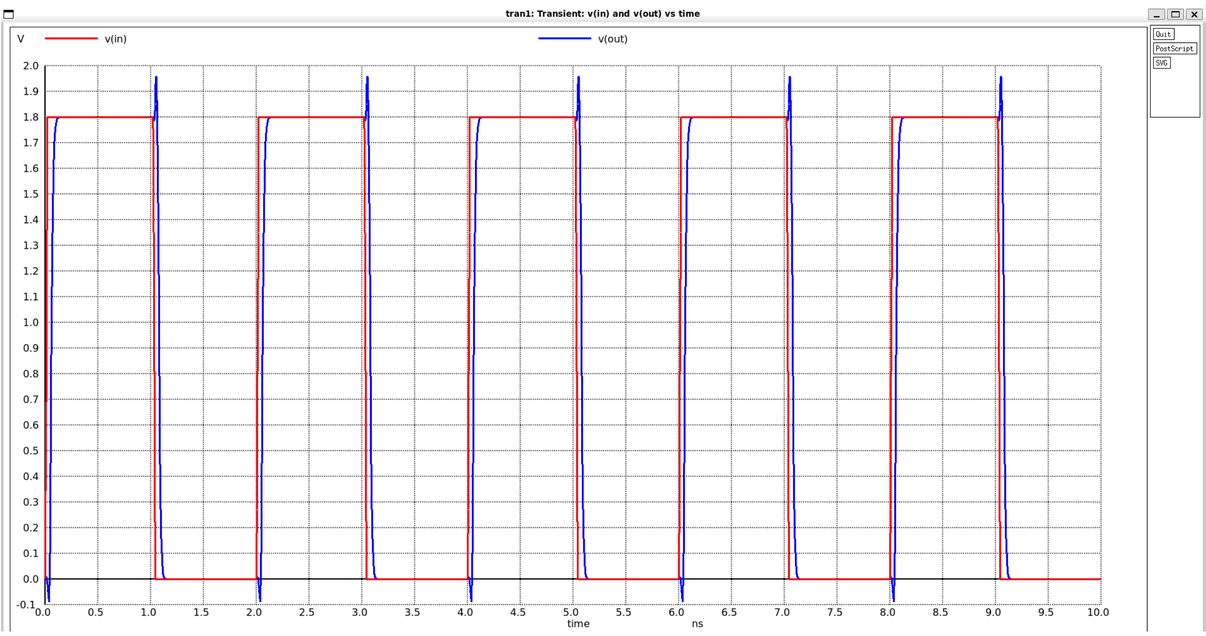
```
No property area found for device sky130_fd_pr__pfet_01v8
No property perim found for device sky130_fd_pr__pfet_01v8
No property topography found for device sky130_fd_pr__pfet_01v8
Model sky130_fd_pr__pfet_01v8 pin 1 == 3
No property mult found for device sky130_fd_pr__pfet_01v8
No property sa found for device sky130_fd_pr__pfet_01v8
No property sb found for device sky130_fd_pr__pfet_01v8
No property sd found for device sky130_fd_pr__pfet_01v8
No property nf found for device sky130_fd_pr__pfet_01v8
No property nrd found for device sky130_fd_pr__pfet_01v8
No property nrs found for device sky130_fd_pr__pfet_01v8
No property area found for device sky130_fd_pr__pfet_01v8
No property perim found for device sky130_fd_pr__pfet_01v8
No property topography found for device sky130_fd_pr__pfet_01v8
Comparison output logged to file comp.out
Logging to file "comp.out" enabled
Circuit sky130_fd_pr__pfet_01v8 contains no devices.
Circuit sky130_fd_pr__nfet_01v8 contains no devices.

Contents of circuit 1:  Circuit: 'copy'
Circuit copy contains 4 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:   2
  Class: sky130_fd_pr__pfet_01v8 instances:   2
Circuit contains 5 nets.
Contents of circuit 2:  Circuit: 'buffer'
Circuit buffer contains 4 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:   2
  Class: sky130_fd_pr__pfet_01v8 instances:   2
Circuit contains 5 nets.

Circuit 1 contains 4 devices, Circuit 2 contains 4 devices.
Circuit 1 contains 5 nets,    Circuit 2 contains 5 nets.


Final result:
Circuits match uniquely.
.
Logging to file "comp.out" disabled
LVS Done.
ankitraj@ANKIT:/mnt/c/Users/ankit/PROJECT1$
```

DRC Clean

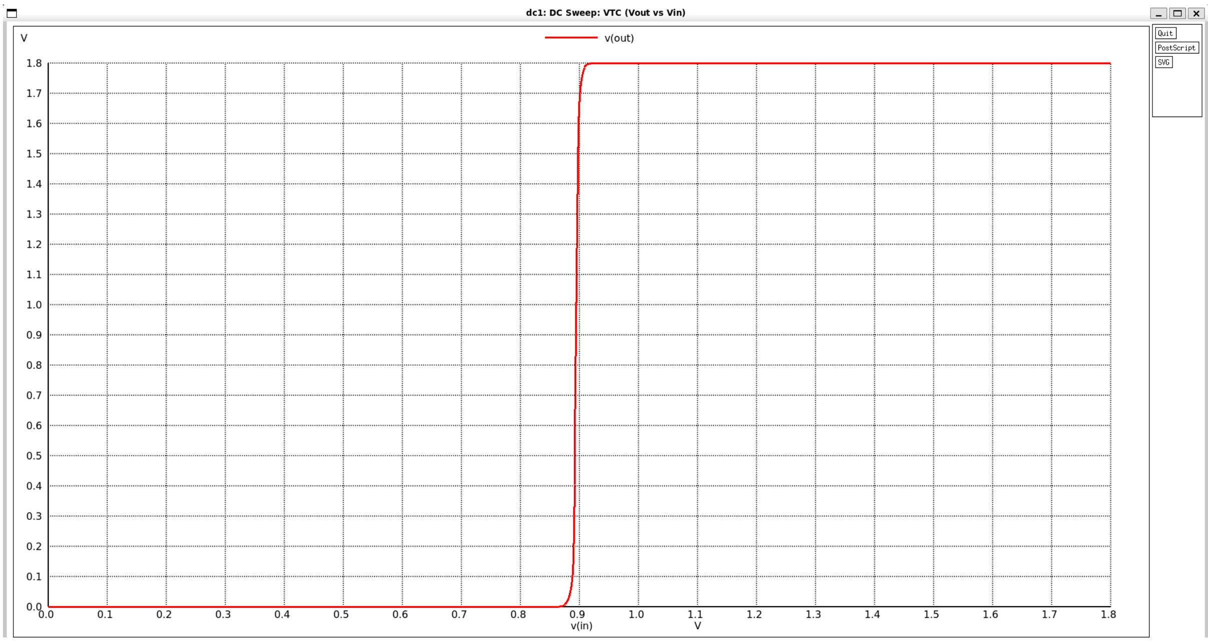




Input output waveforms which were used to rise, fall, and propagation delay.

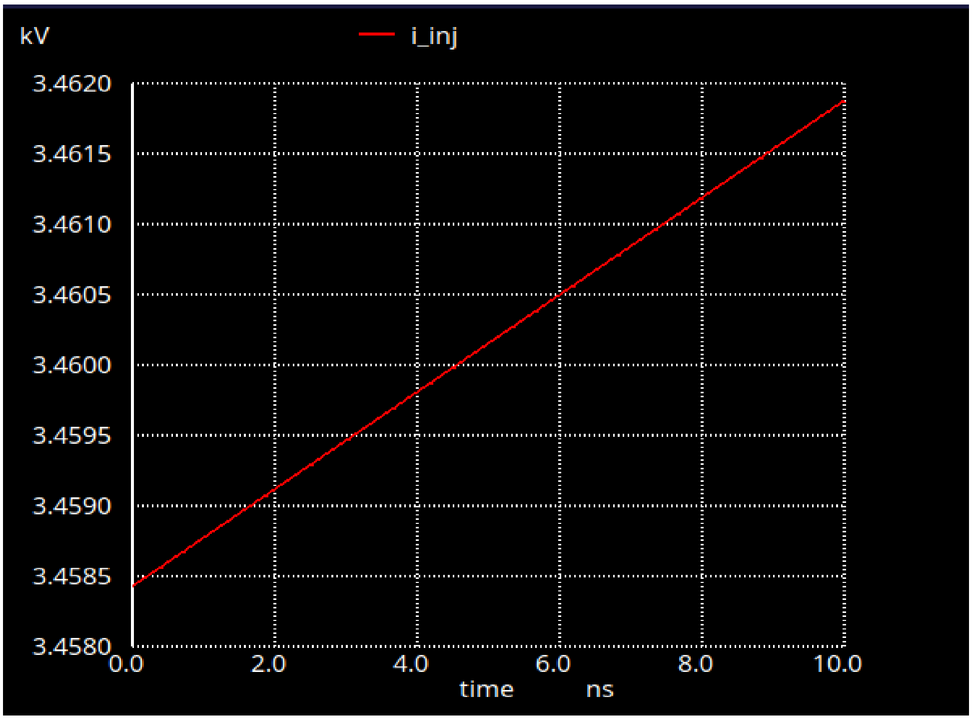


Vout vs Vin

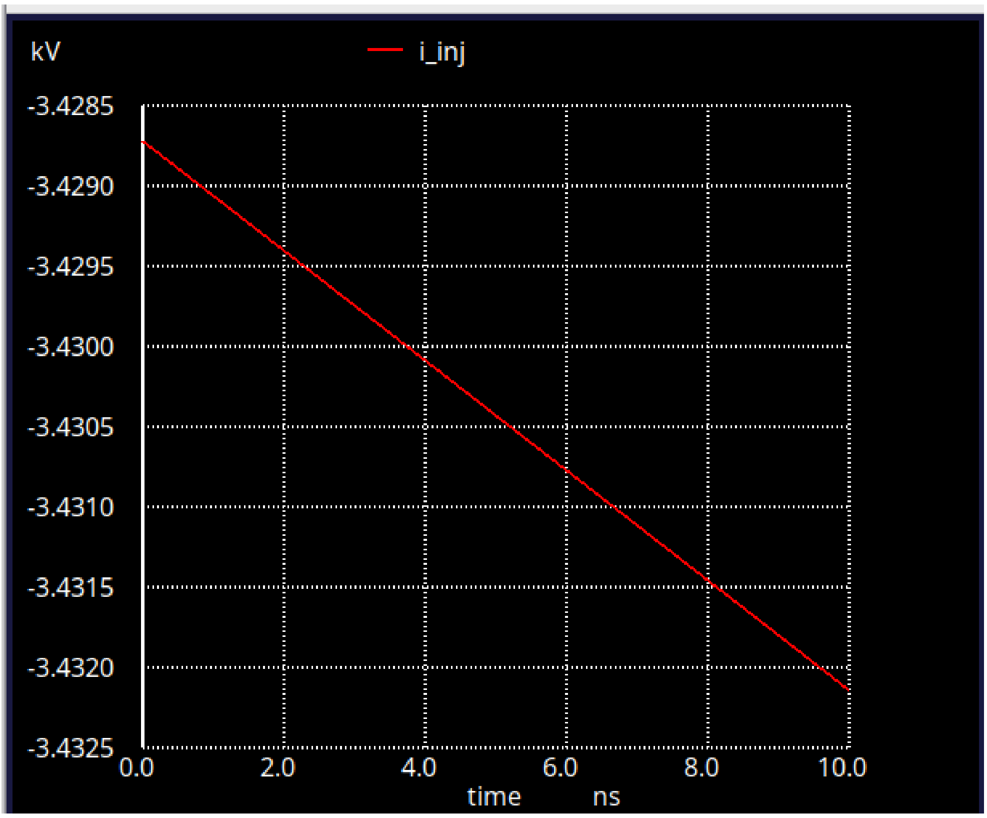



```
Initial Transient Solution
--------------------------

Node                                   Voltage
----                                   -------
vdd                                        1.8
in                                           0
xbuffer.out1                               1.8
out                                1.17551e-07
out2                                       1.8
v1#branch                                    0
vdd#branch                        -6.64692e-11


No. of Data Rows : 1068
Warning: Missing charsets in String to FontSet conversion
final_rise          =  2.202640e-11 targ=  7.043509e-11 trig=  4.840869e-11
final_fall          =  2.206229e-11 targ=  1.090007e-09 trig=  1.067945e-09
Rise time at optimum: 22.0264 ps
Fall time at optimum: 22.0623 ps
tphl                =  1.067889e-09 targ=  1.077889e-09 trig=  1.000000e-11
tplh                =  -9.720107e-10 targ=  5.798928e-11 trig=  1.030000e-09
 tp_PARAM = 47.9392 ps
vih                 =  9.133838e-01
vil                 =  8.721029e-01
vm                  =  8.938622e-01
vol                 =  1.175511e-07
voh                 =  1.800000e+00
 NMH_val = 0.886616   NML_val=0.872103
ngspice 1 ->
```

Input capacitance was measured by applying current source of 1uA at the Gate
Now we CV = It
Therefore, the slope of Voltage(V) vs time(t) graph gives I/C.
I is fixed at 1 uA.
Therefore C can be obtained as I/(Slope of V vs t graph)



Slope = 3.450724e+08, C = i/slope, therefore rise cap = (1e-06)/(3.4750724e+08) = 0.0028978 pF



Slope = -3.42237e+08, C = i/slope , fall capacitance = (-1e-06)/(-3.42237e+08) = 0.00292195 pF

a) Input pin capacitances:

Input Pins	Rise Cap (pF)	Fall Cap (pF)	Average Cap (pF)
A	0.0028979	0.00292195	0.00292195

b) **Transition Time Table:** (please strictly consider 20% and 80% of VDD for transition time)

(i) Output Rise Transitions (in ns) [Input slew vs output capacitance].

Related pin A:

	10 ps	100 ps	1000 ps
0.5 fF	0.02440	0.01896	0.03477
10 fF	0.68230	0.06873	0.07696
100 fF	0.59246	0.59248	0.59306

(ii) Output Fall Transitions (in ns) [Input slew vs output capacitance].

Related pin A:

	10 ps	100 ps	1000 ps
0.5 fF	0.22510	0.01793	0.32600
10 fF	0.74120	0.07377	0.08190
100 fF	0.64161	0.64165	0.64250

c) **Propagation delay time tables**: (unlike textbook definitions that we used for our assignments, here we will use 50% of input to 50% of output to simulate propagation delay – by keeping other inputs fixed).

(i) Cell Rise Delay (in ns) [Input slew vs output capacitance].

Related pin A:

	10 ps	100 ps	1000 ps
0.5 fF	0.04929	0.06676	0.14594
10 fF	0.08926	0.10664	0.19889
100 fF	0.44732	0.46460	0.55730

(ii) Cell Fall Delay (in ns) [Input slew vs output capacitance].

Related pin A:

	10 ps	100 ps	1000 ps
0.5 fF	0.04688	0.06883	0.15840
10 fF	0.09671	0.11781	0.21900
100 fF	0.53255	0.55408	0.65790

d) **Static Power (all possible input combinations based on number of inputs(1 here)).**

Condition (A)	Power (nW)
0	0.1163
1	0.1163

e) Dynamic Power Table:

(i) Rise Power (in nW) [Input slew vs output capacitance].

Related pin A

	10 ps	**100 ps**	**1000 ps**
0.5 fF	339.62	361.63	364.24
10 fF	1965.93	1972.5	2160.98
100 fF	18111.4	18106.5	17994

(ii) Fall Power (in nW) [Input slew vs output capacitance].

Related pin A

	10 ps	**100 ps**	**1000 ps**
0.5 fF	197.13	212.47	247.94
10 fF	170.06	173.89	243.33
100 fF	171.46	170.74	173.19

Dynamic power calculation ngspice netlist screenshot

```
X1 A VDD 0 Y copy
C1 Y 0 100f

Vin A 0 PULSE(0 1.8 0 1000p 1000p 3n 6n)
Vdd VDD 0 1.8

.tran 1ps 20ns 0 5ps

.control
run
meas tran tr20 when v(Y)=0.2*1.8 rise=1
meas tran tr80 when v(Y)=0.8*1.8 rise=1
meas tran curr_int_r integ vdd#branch from=tr20 to=tr80
let power_int_r=curr_int_r*1.8
print power_int_r/6e-09
meas tran tf80 when v(Y)=0.8*1.8 fall=1
meas tran tf20 when v(Y)=0.2*1.8 fall=1
meas tran curr_int_f integ vdd#branch from=tf80 to=tf20
let power_int_f=curr_int_f*1.8
print power_int_f/6e-09
.endc
.end
```

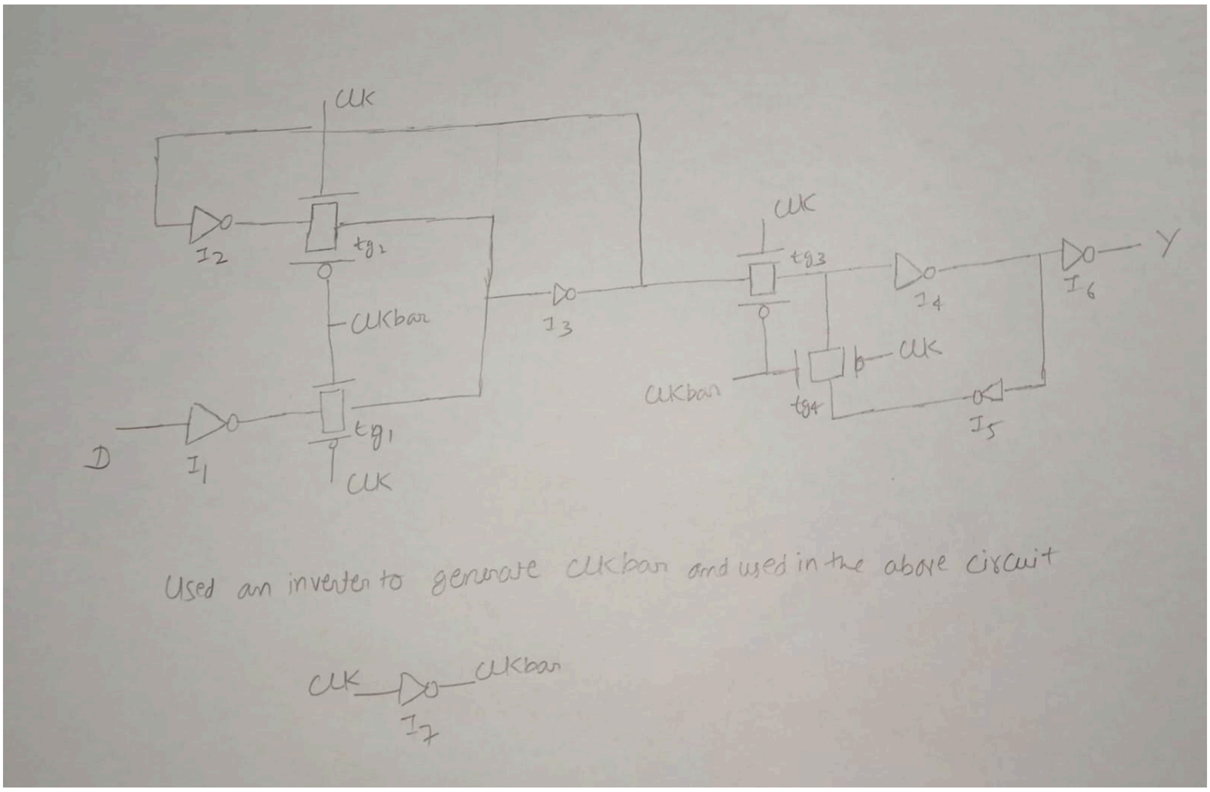
Static power calculation ngspice netlist screenshot

```
X1 A VDD 0 Y copy
C1 Y 0 0.5f
Vdd VDD 0 1.8
Vi A 0 0

.control
run
*dc Vi 0 0 1.8
dc Vi 0 1.8 1.8
print vdd#branch
let power_static=vdd#branch*vdd
print power_static
.endc
.end
```

Verilog code compiled successfully.

```
newuser@harsh:~/Project1/Team Number 23/Verilog$ iverilog buf.v
newuser@harsh:~/Project1/Team Number 23/Verilog$
```

Positive Edge Flip Flop(dfxtp)

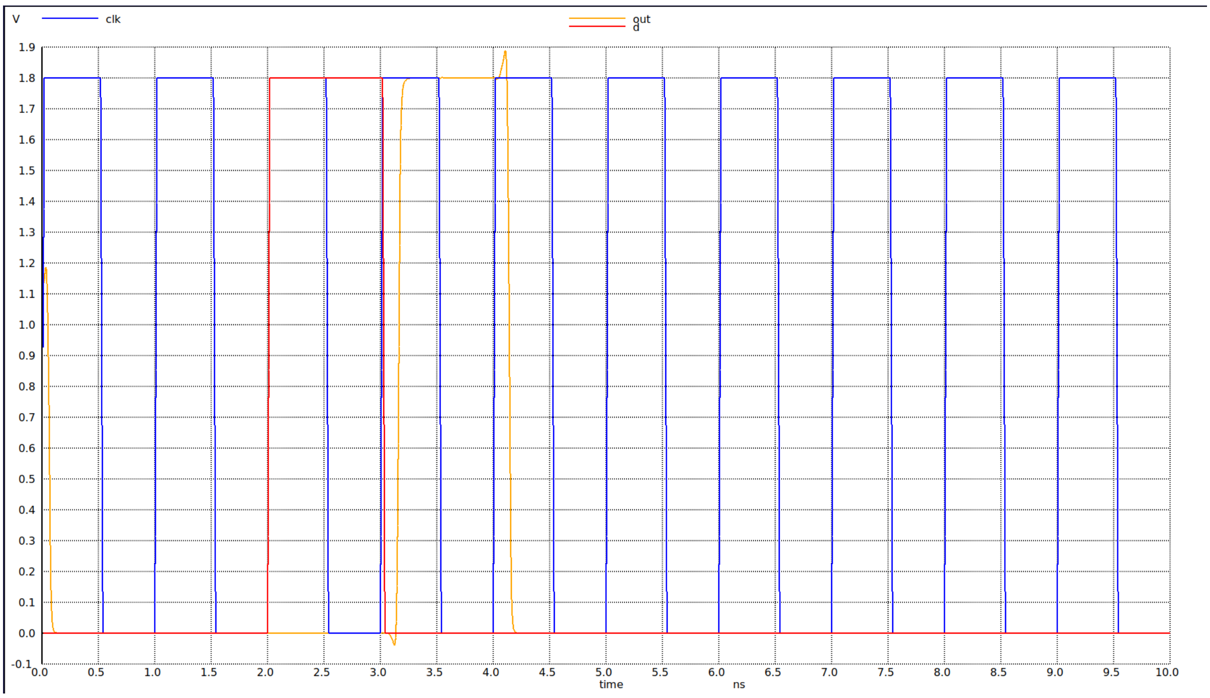


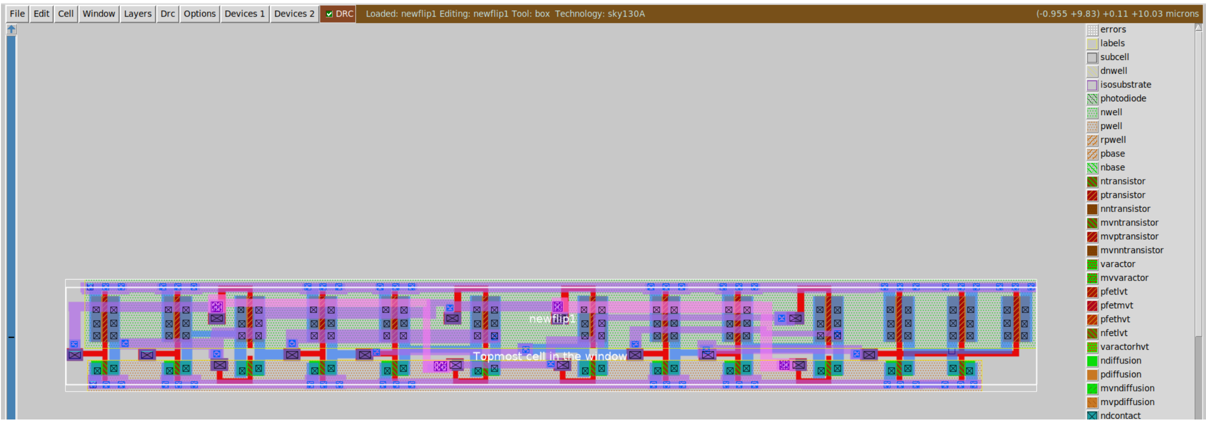
	Wn(in µm)	Wp(in µm)	L(in µm)
I1	0.42	1.27	0.15
I2	0.42	1.27	0.15
I3	0.42	1.27	0.15
I4	0.42	1.27	0.15
I5	0.42	1.27	0.15
I7	0.42	1.27	0.15
TG1	0.42	1.27	0.15
TG2	0.42	1.27	0.15
TG3	0.42	1.27	0.15
TG4	0.42	1.27	0.15
I6	2*0.42	3*1.27	0.15

Table for the sizes of the pmos and nmos

Ngspice simulation

```
Initial Transient Solution
--------------------------

Node                                       Voltage
----                                       -------
vdd                                            1.8
d                                                0
clk                                              0
xd.clkbar                                      1.8
xd.xmaster.p                                   1.8
xd.xmaster.m                                   1.8
xd.c                                   1.22328e-07
xd.xmaster.o                                   1.8
xd.xslave.m                               0.893863
xd.xslave.n                               0.893863
xd.xslave.o                               0.893863
out                                        1.13585
out1                                     0.0102219
v2#branch                                        0
v1#branch                                        0
vdd#branch                             -5.76456e-05

 Reference value :  9.21350e-09
No. of Data Rows : 10111
Warning: Missing charsets in String to FontSet conversion
final_rise          =  2.147469e-11 targ=  3.174226e-09 trig=  3.152752e-09
final_fall          =  2.180685e-11 targ=  4.154651e-09 trig=  4.132845e-09
Rise time at optimum: 21.4747 ps
Fall time at optimum: 21.8068 ps
ngspice 1 ->
```

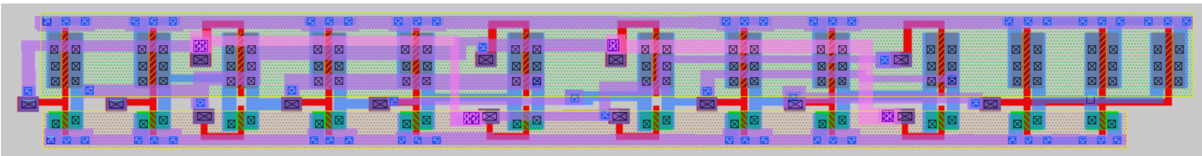




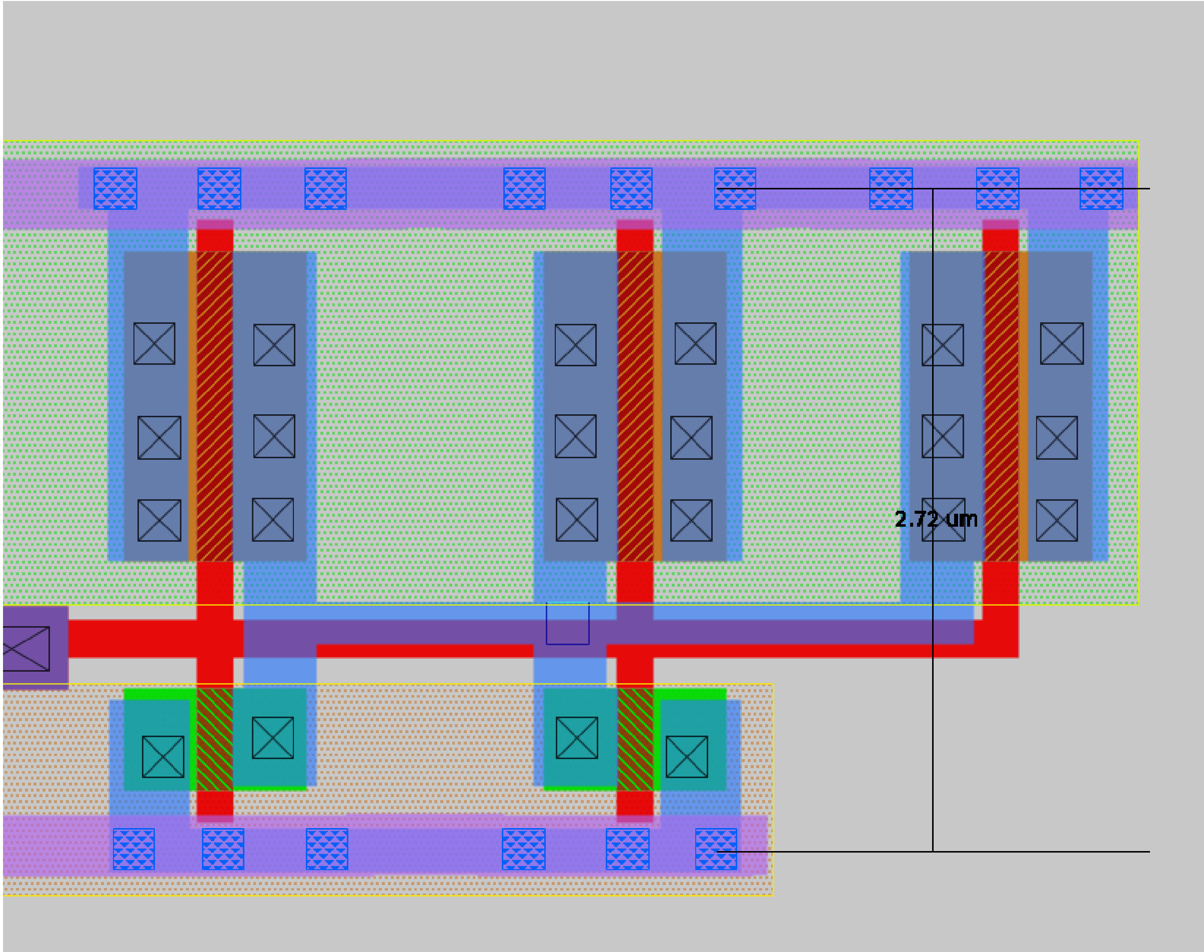
DRC Clean

```
Root cell box:
           width x height  (   llx,  lly  ), (   urx,  ury  )  area (units^2)

microns:  26.780 x 2.720   (-1.065, -0.200), ( 25.715,  2.520)  72.842
lambda:   2678.00 x 272.00  (-106.50, -20.00), ( 2571.50,  252.00)  728416.00
internal:   5356 x 544     (  -213, -40   ), (  5143,  504  )  2913664
% |
```

Box Area



Dfip Flop



Measure

LVS Mapping

```
Comparison output logged to file comp.out
Logging to file "comp.out" enabled
Circuit sky130_fd_pr__pfet_01v8 contains no devices.
Circuit sky130_fd_pr__nfet_01v8 contains no devices.

Contents of circuit 1:  Circuit: 'newflip1'
Circuit newflip1 contains 25 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:  12
  Class: sky130_fd_pr__pfet_01v8 instances:  13
Circuit contains 13 nets.
Contents of circuit 2:  Circuit: 'dflip'
Circuit dflip contains 22 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:  11
  Class: sky130_fd_pr__pfet_01v8 instances:  11
Circuit contains 13 nets.

Circuit was modified by parallel/series device merging.
New circuit summary:

Contents of circuit 1:  Circuit: 'newflip1'
Circuit newflip1 contains 22 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:  11
  Class: sky130_fd_pr__pfet_01v8 instances:  11
Circuit contains 13 nets.
Contents of circuit 2:  Circuit: 'dflip'
Circuit dflip contains 22 device instances.
  Class: sky130_fd_pr__nfet_01v8 instances:  11
  Class: sky130_fd_pr__pfet_01v8 instances:  11
Circuit contains 13 nets.

Circuit 1 contains 22 devices, Circuit 2 contains 22 devices.
Circuit 1 contains 13 nets,    Circuit 2 contains 13 nets.


Final result:
Circuits match uniquely.
.
Logging to file "comp.out" disabled
LVS Done.
ankitraj@ANKIT:/mnt/c/Users/ankit$
```

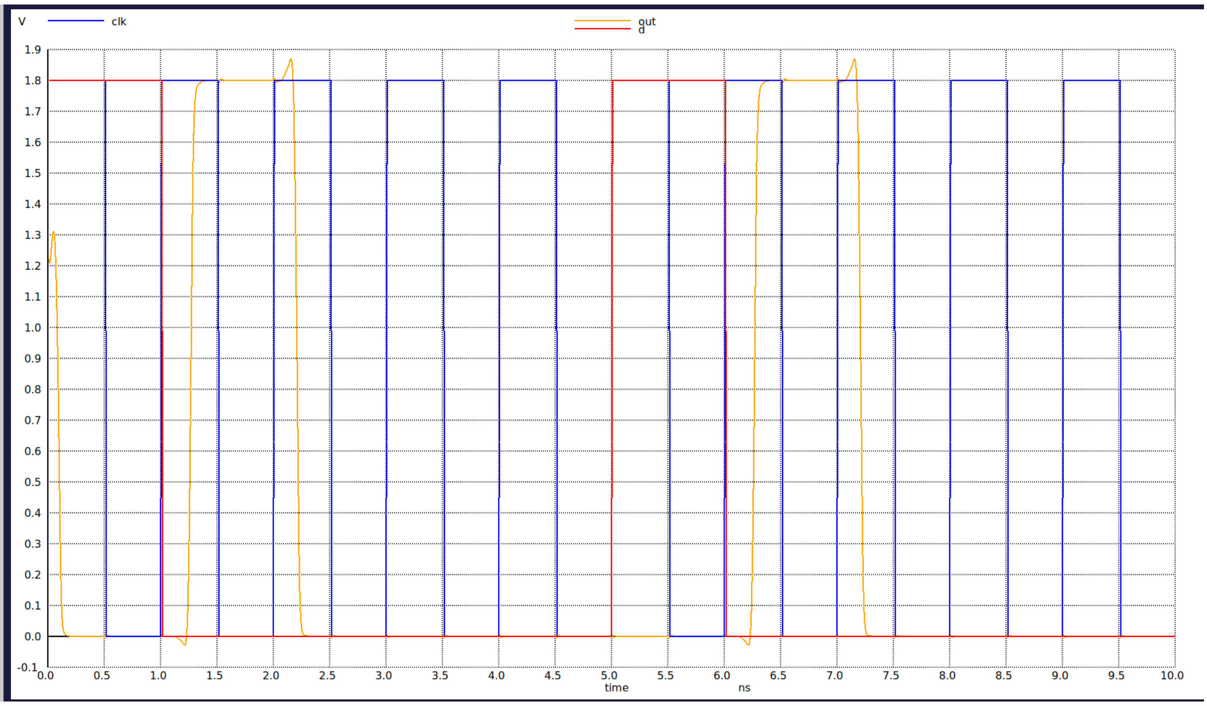
Pex Extraction

```
* NGSPICE file created from newflip1.ext - technology: sky130A

.subckt newflip1 a Q clk vdd vss
X0 a_20_10# clk vdd vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X1 a_3512_10# a_3112_10# vdd vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X2 vdd a_3112_10# Q vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X3 a_820_10# a_20_10# a_420_10# vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X4 a_2712_10# a_20_10# a_3512_10# vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X5 a_420_10# a vss vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X6 a_420_10# a vdd vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X7 a_2712_10# clk a_1220_10# vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X8 Q a_3112_10# vss vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X9 a_3112_10# a_2712_10# vdd vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X10 vdd a_3112_10# Q vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X11 Q a_3112_10# vdd vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X12 a_3112_10# a_2712_10# vss vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X13 a_3512_10# a_3112_10# vss vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X14 a_2712_10# clk a_3512_10# vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X15 a_1620_10# a_1220_10# vdd vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X16 a_2712_10# a_20_10# a_1220_10# vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X17 a_20_10# clk vss vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X18 a_1220_10# a_820_10# vdd vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X19 a_820_10# a_20_10# a_1620_10# vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X20 a_820_10# clk a_420_10# vdd sky130_fd_pr__pfet_01v8 ad=0.381 pd=3.14 as=0.381 ps=3.14 w=1.27 l=0.15
**devattr s=15240,628 d=15240,628
X21 a_820_10# clk a_1620_10# vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X22 vss a_3112_10# Q vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X23 a_1220_10# a_820_10# vss vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
X24 a_1620_10# a_1220_10# vss vss sky130_fd_pr__nfet_01v8 ad=0.126 pd=1.44 as=0.126 ps=1.44 w=0.42 l=0.15
**devattr s=5040,288 d=5040,288
```

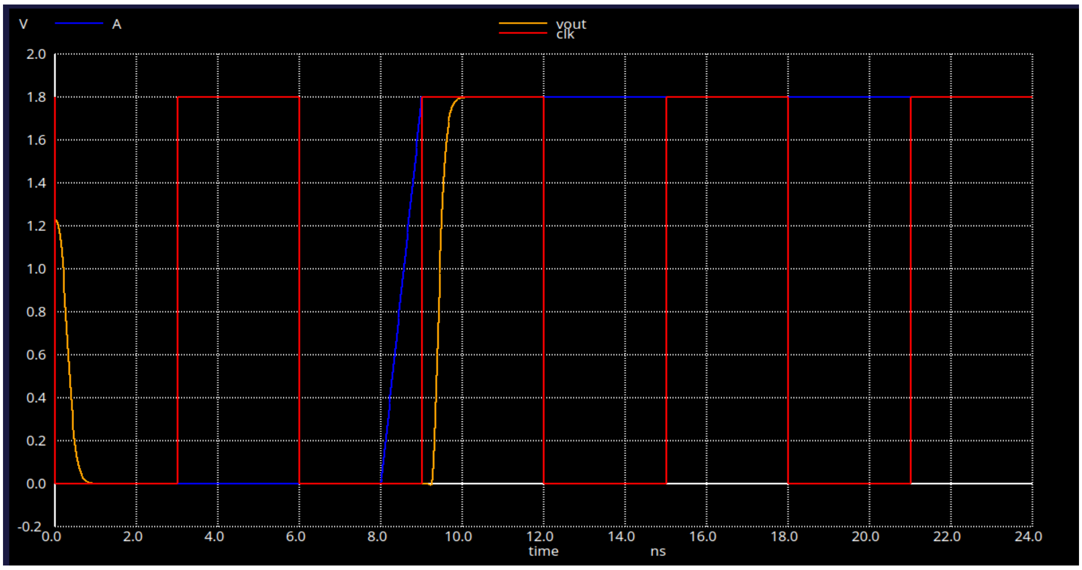
```
C0 clk a_3512_10# 0.14043f
C1 vdd clk 2.03253f
C2 clk a_1220_10# 0.67349f
C3 Q a_20_10# 0.00416f
C4 a_1620_10# a_820_10# 0.29214f
C5 Q a_2712_10# 0.00409f
C6 vdd a_3512_10# 0.26273f
C7 vdd a_1220_10# 0.44813f
C8 a a_820_10# 0.00139f
C9 a_20_10# a_3112_10# 0.18541f
C10 a a_420_10# 0.03889f
C11 a_2712_10# a_3112_10# 0.53129f
C12 a_1620_10# clk 0.19621f
C13 a clk 0.02194f
C14 vdd a_1620_10# 0.25504f
C15 a_20_10# a_2712_10# 0.1427f
C16 a_1620_10# a_1220_10# 0.13205f
C17 a vdd 0.11687f
C18 Q clk 0
C19 a_3112_10# clk 0.04901f
C20 Q vdd 0.66603f
C21 a_20_10# a_820_10# 0.64225f
C22 a_2712_10# a_820_10# 0
C23 a_20_10# a_420_10# 0.2848f
C24 a_3112_10# a_3512_10# 0.16752f
C25 vdd a_3112_10# 0.7595f
C26 a_3112_10# a_1220_10# 0
C27 a_20_10# clk 1.04469f
C28 a_2712_10# clk 0.64975f
C29 a_420_10# a_820_10# 0.1755f
C30 a_20_10# a_3512_10# 0.16547f
C31 a_20_10# vdd 2.14675f
C32 a_20_10# a_1220_10# 0.20529f
C33 a_2712_10# a_3512_10# 0.25537f
C34 vdd a_2712_10# 0.45124f
C35 a_2712_10# a_1220_10# 0.17594f
C36 clk a_820_10# 0.19632f
C37 a_420_10# clk 0.20341f
C38 vdd a_820_10# 0.39027f
C39 a_820_10# a_1220_10# 0.57214f
C40 a_420_10# vdd 0.2362f
C41 a_20_10# a_1620_10# 0.18068f
C42 Q a_3112_10# 0.1733f
C43 a_420_10# a_1220_10# 0
C44 a_2712_10# a_1620_10# 0
C45 a a_20_10# 0.05767f
C46 Q vss 0.30901f
C47 vdd vss 7.96177f
C48 a vss 0.25781f
C49 clk vss 2.15927f
C50 a_3512_10# vss 0.21418f
C51 a_3112_10# vss 1.31837f
C52 a_2712_10# vss 0.4868f
C53 a_1620_10# vss 0.19334f
C54 a_1220_10# vss 0.63554f
C55 a_820_10# vss 0.51092f
C56 a_420_10# vss 0.18235f
C57 a_20_10# vss 1.64347f
.ends
```

Pex Simulation

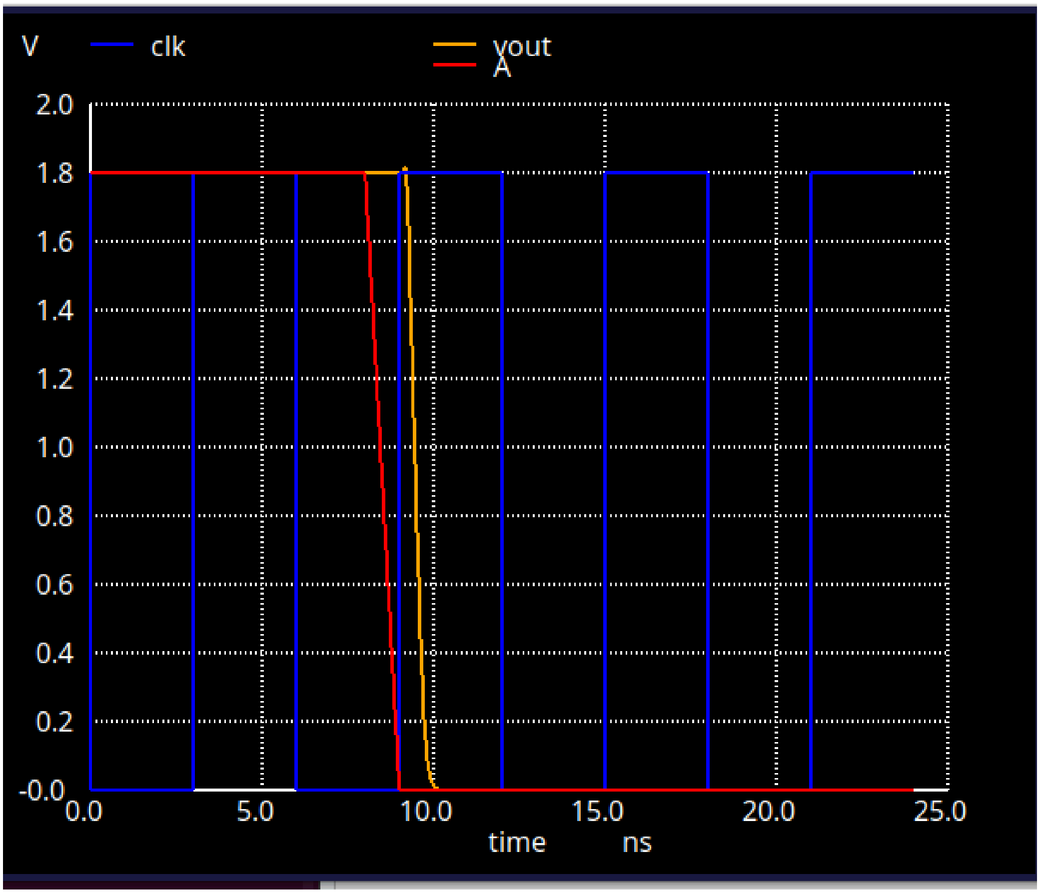
```
Initial Transient Solution
--------------------------

Node                                    Voltage
----                                    -------
vdd                                         1.8
d                                             0
clk                                           0
out                                     1.22862
out1                                  0.0023024
v2#branch                                     0
v1#branch                                     0
vdd#branch                         -5.56138e-05

 Reference value :  9.07450e-09
No. of Data Rows : 10114
Warning: Missing charsets in String to FontSet conversion
final_rise          =  3.025394e-11 targ=  1.284676e-09 trig=  1.254422e-09
final_fall          =  3.135961e-11 targ=  2.225230e-09 trig=  2.193871e-09
Rise time at optimum: 30.2539 ps
Fall time at optimum: 31.3596 ps
```



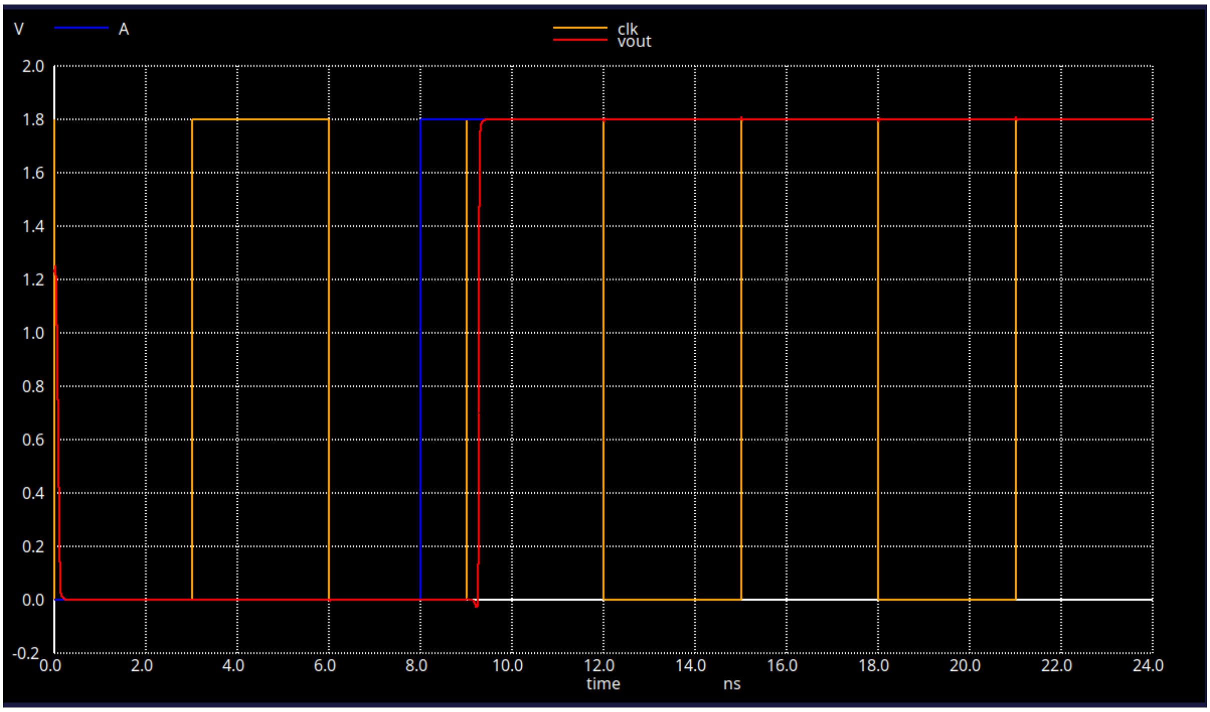
One of the graph used to obtain Rise transition time of vout



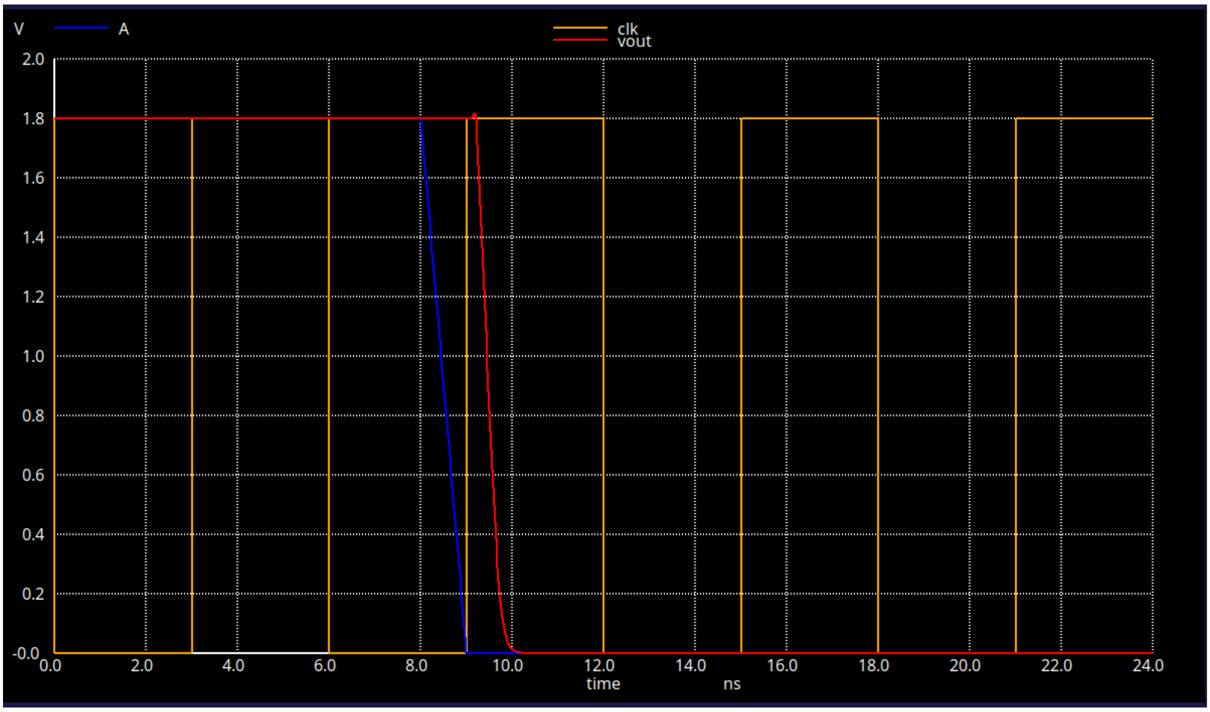
One graph used to calculate fall transition time of vout



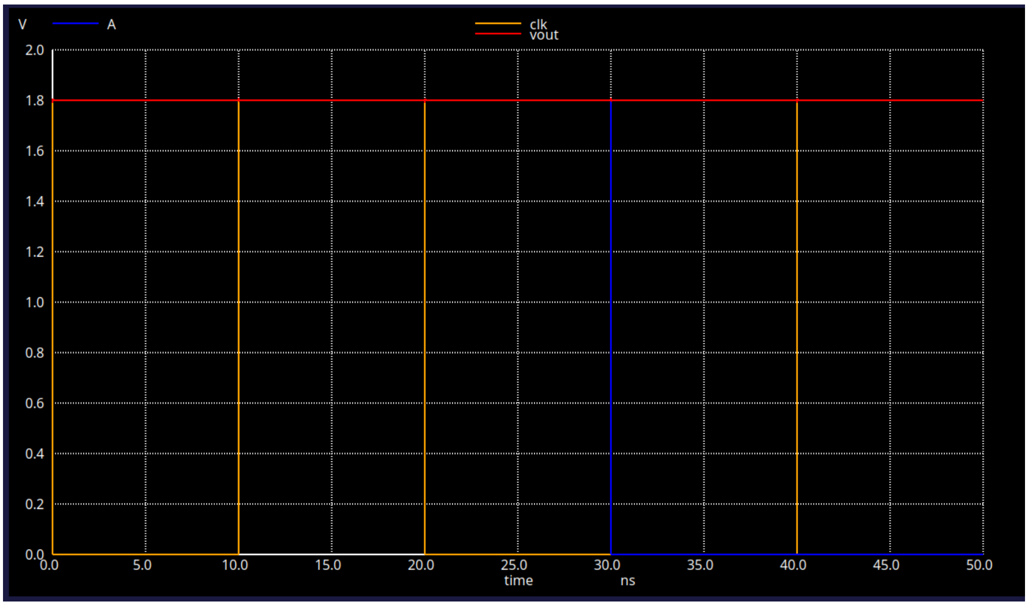
One of the graph used to calculate clk to Q(vout) delay rise transition



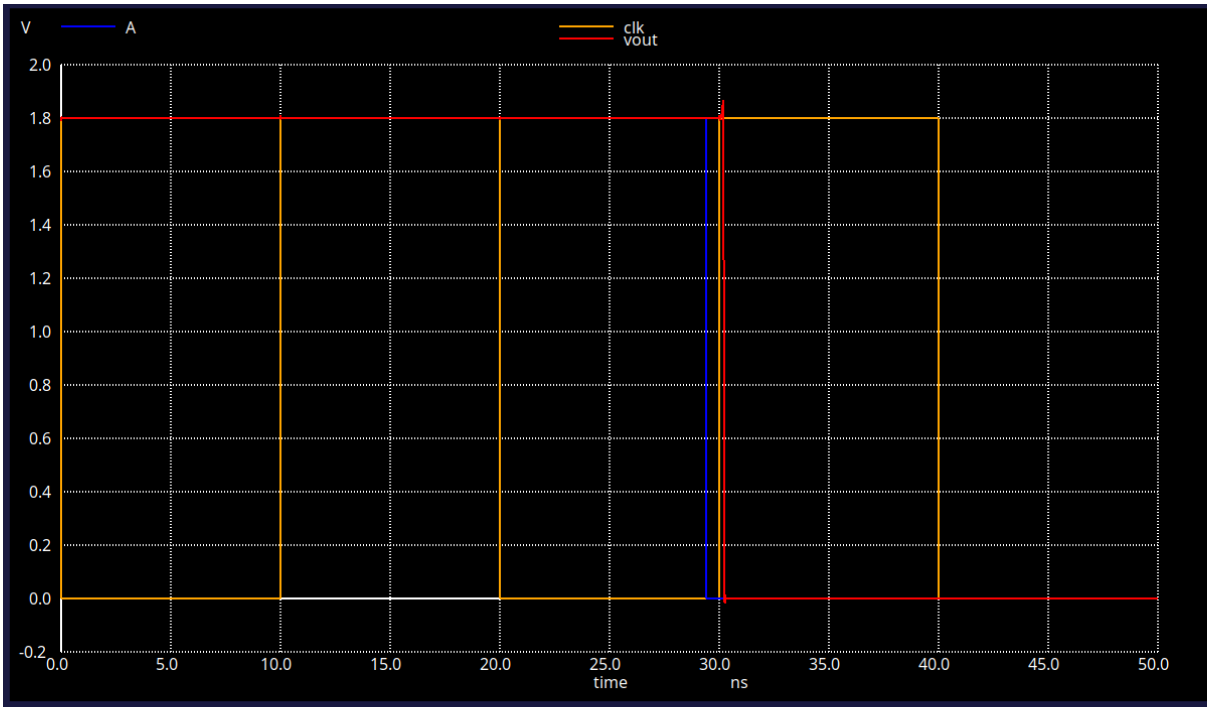
One of the graph used to calculate Clk to Q(vout) delay during fall transition



Below waveform shows setup time violation



Below graph shows waveform setup time is not violated



Below is the reference code which we used for calculating setup time

```
** Transition Testing Netlist
.param delta=200p

.include dfxtp_pex.spice
.lib /usr/local/share/pdk/sky130A/libs.tech/ngspice/sky130.lib.spice tt

x1 A vout clk vdd GND dfxtp
V1 A GND PWL(0 1.8 6n 1.8 {30n - delta} 1.8 {30n + 10p - delta} 0)
VCLK clk GND PULSE(1.8 0 0 1000p 1000p 10n 20n)
V3 vdd GND 1.8

.tran 0.02n 50n

.control
foreach delt  10p 7p 5p 4p 3p 2p
    alterparam delta=$delt
    reset
    run
    setplot tran
    let delp=$delt
    meas tran qval find v(vout) at=40n
    if qval < 0.9
        echo "not"
        print delp
  end
end
.endc

.save all
.end
.GLOBAL GND
.end
```

For Sequential cells, the following characterizations have to be performed and filled.

a. Input pin capacitances:

Input Pins	Rise Cap (pF)	Fall Cap (pF)	Average Cap (pF)
D	0.00216453	0.00217269	0.00216861
CLK	0.00520604	0.00485335	0.0050296

b. Set-up Time Table:

(i) Rise Constraint (in ns) [Input slew vs CLK slew].

	10 ps	**1000 ps**
10 ps	0.110	0.800
1000 ps	0.002	0.002

(ii) Fall Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0.150	0.600
1000 ps	0.002	0.002

c. Hold Time Table:

(i) Rise Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0*	0*
1000 ps	0*	0*

(ii) Fall Constraint (in ns) [Input slew vs CLK slew].

	10 ps	1000 ps
10 ps	0*	0*
1000 ps	0*	0*

*By theory we know that schematic we used for DFF have 0 hold time

d. Transition Time Table (for Q output, CLK will have minimum slew of 10 ps): (please strictly consider 20% and 80% of VDD for transition time)

(i) Output Rise Transitions (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.03119	0.03119	0.03264
10 fF	0.04830	0.04830	0.04766
100 fF	0.20622	0.20622	0.20620

(ii) Output Fall Transitions (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.03081	0.03081	0.03043
10 fF	0.05773	0.05773	0.05663
100 fF	0.33051	0.33051	0.33057

e. CLK-to-Q Delay Time Table: (delay between clock transition and data transition. Use 50% of CLK to 50% of output to simulate propagation delay).

(i) Cell Rise Delay (in ns) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	100 ps	1000 ps
0.5 fF	0.25495	0.25495	0.25438
10 fF	0.28025	0.28025	0.28014

100 fF	0.41035	0.41035	0.41039

(ii) Cell Fall Delay (in ns) [Input slew vs output capacitance].
Related pin D: (i.e., other input pins are held constant)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.20015	0.20015	0.20030
10 fF	0.23393	0.23393	0.23404
100 fF	0.45925	0.45925	0.45943

f. Static Power (<u>all possible input combinations</u> of CLK and D).

Condition (CLK, D)	**Power (nW)**
00	**98870**
01	**98870**
10	**144800**
11	**144800**

g. Dynamic Power Table: (CLK will have minimum slew of 10 ps)

(i) Rise Power (in nW) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	**11461.50**	**11460.52**	**14456.67**
10 fF	**220.91**	**224.18**	**308.96**
100 fF	**2286.14**	**2688.61**	**3197.86**

(ii) Fall Power (in nW) [Input slew vs output capacitance].

Related pin D: (i.e., other input pins are held constant)

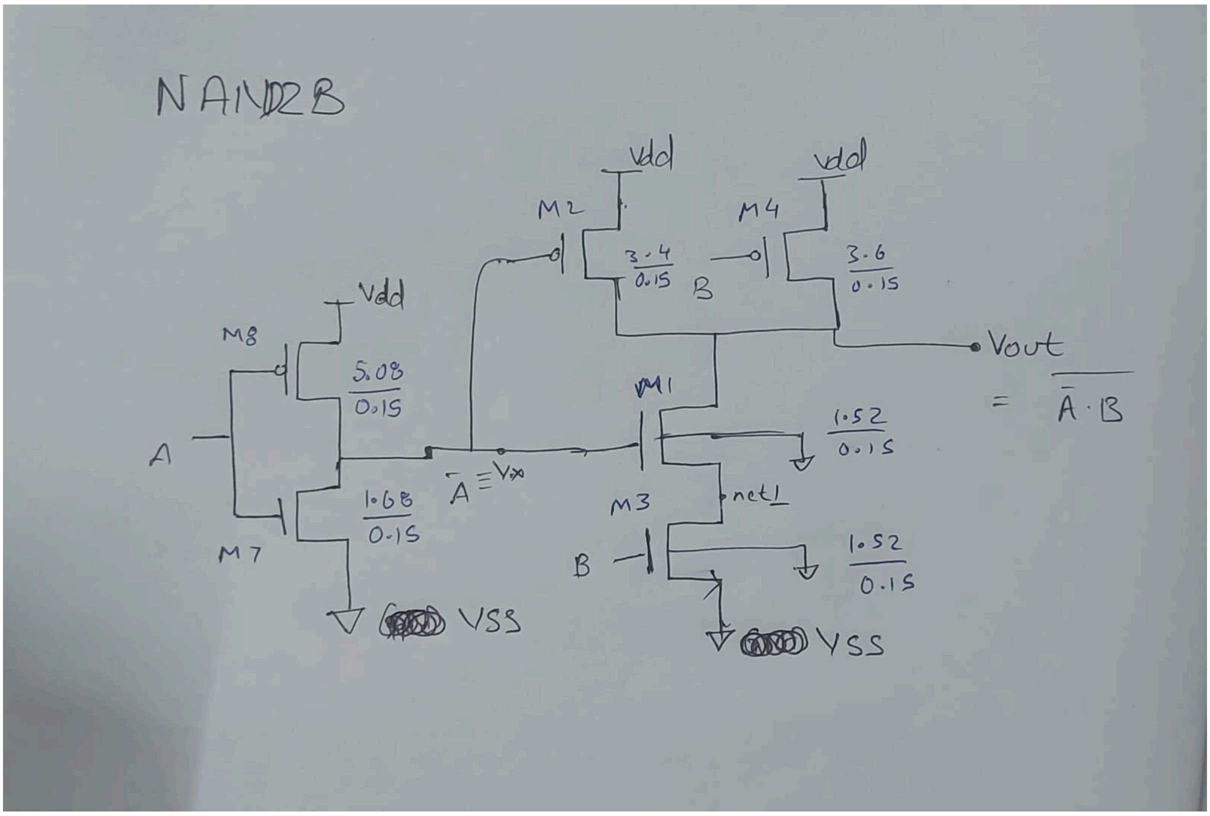
	10 ps	**100 ps**	**1000 ps**
0.5 fF			
10 fF			
100 fF			

D Flip Flop

```
newuser@harsh:~/Project1/dff$ iverilog dff.v
newuser@harsh:~/Project1/dff$
```

Nand2b

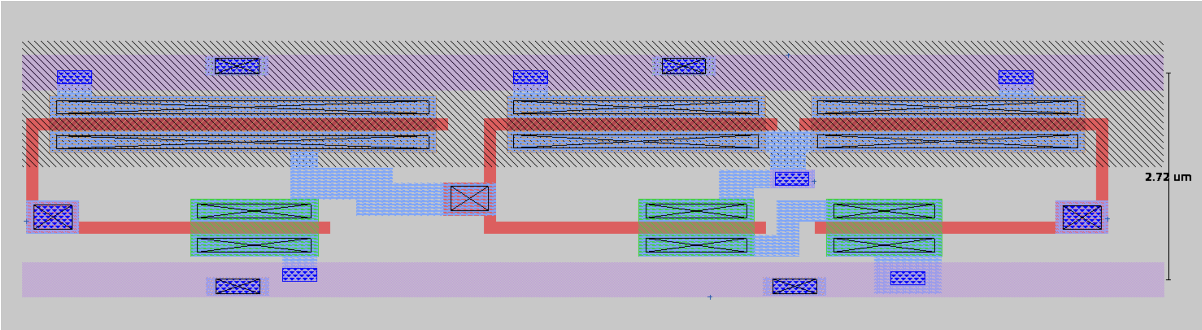
Circuit Diagram

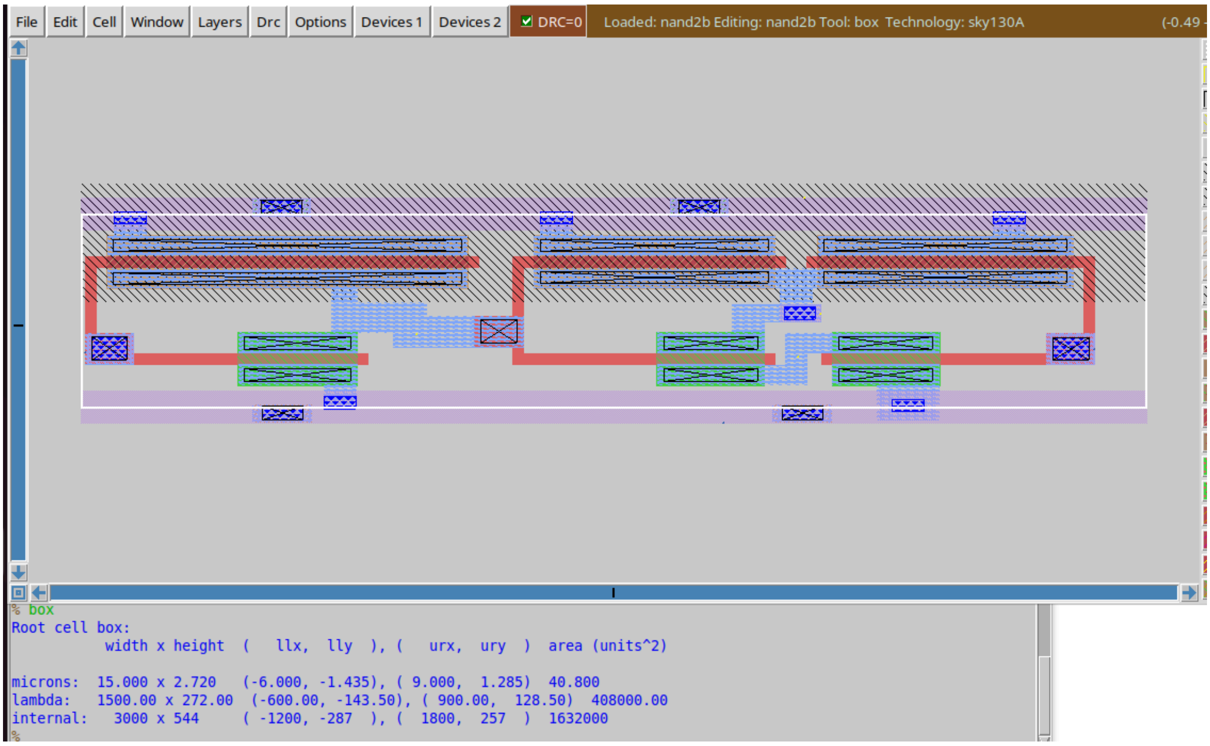


	W(in um)	L(in um)
M1	1.52	0.15
M2	3.4	0.15
M3	1.52	0.15
M4	3.6	0.15
M7	1.68	0.15
M8	5.08	0.15

The sizing of transistor was done to match the rise time and fall time of INVX1 that is 19ps
Final Rise and Fall time with the sizing given above when INVX1 is load to NAND2B gate
For Input A when B is kept at VDD
tr(rise time) = 19.5 ps
tf(fall time) = 20.8 ps
For input B when A is kept at GND
tr = 19.22 ps
tf = 19.32 ps

Layout Screenshot





Cell area = width x height = 15um x 2.27um

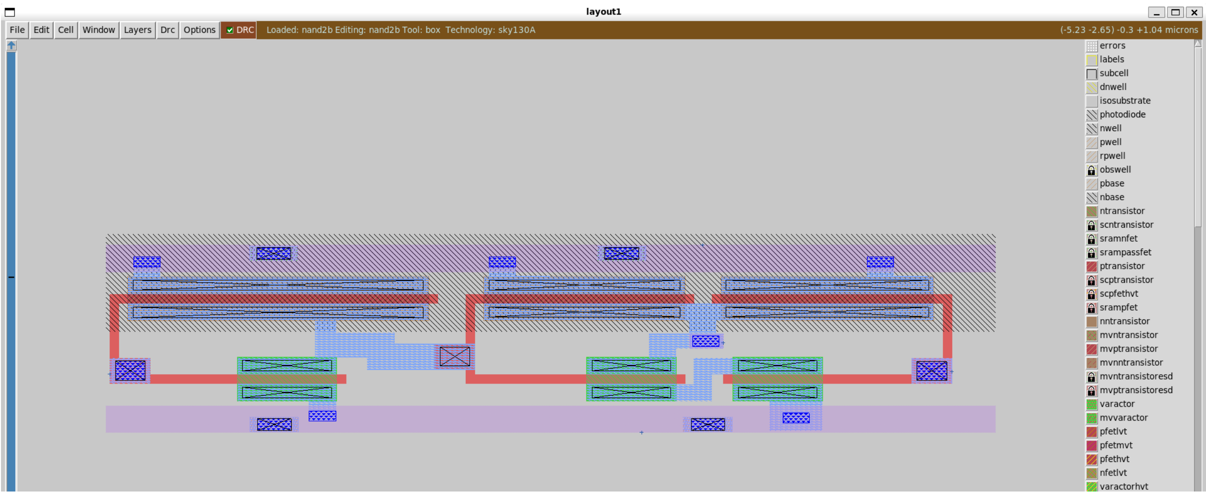
Pex netlist screenshot

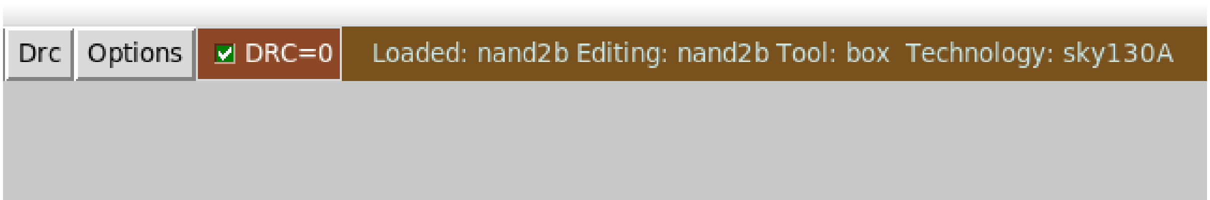
```
* NGSPICE file created from nand2b_mod.ext - technology: sky130A
.subckt A B VDD VSS Y
X0 VDD A a_n564_24# VDD sky130_fd_pr__pfet_01v8 ad=1.524 pd=10.91695 as=1.524 ps=10.76 w=5.08 l=0.15
X1 a_210_n112# B VSS VSS sky130_fd_pr__nfet_01v8 ad=0.456 pd=3.64 as=0.456 ps=3.61 w=1.52 l=0.15
X2 VDD B Y VDD sky130_fd_pr__pfet_01v8 ad=1.08 pd=7.73642 as=1.08 ps=7.81714 w=3.6 l=0.15
X3 a_n564_24# A VSS VSS sky130_fd_pr__nfet_01v8 ad=0.504 pd=3.96 as=0.504 ps=3.99 w=1.68 l=0.15
X4 VDD a_n564_24# Y VDD sky130_fd_pr__pfet_01v8 ad=1.02 pd=7.30662 as=1.02 ps=7.38286 w=3.4 l=0.15
X5 Y a_n564_24# a_210_n112# VSS sky130_fd_pr__nfet_01v8 ad=0.456 pd=3.64 as=0.456 ps=3.64 w=1.52 l=0.15
C0 B a_n564_24# 0.01484f
C1 a_n564_24# a_210_n112# 0.02322f
C2 B VDD 0.08462f
C3 VDD a_210_n112# 0.01019f
C4 Y B 0.06632f
C5 Y a_210_n112# 0.25883f
C6 A a_n564_24# 0.11125f
C7 A VDD 0.1025f
C8 Y A 0
C9 a_n564_24# VDD 0.58781f
C10 Y a_n564_24# 0.07831f
C11 Y VDD 0.73699f
C12 B a_210_n112# 0.03056f
C13 A a_210_n112# 0
C14 Y VSS 0.29981f
C15 B VSS 0.61283f
C16 A VSS 0.63095f
C17 VDD VSS 4.16387f
C18 a_210_n112# VSS 0.39452f
C19 a_n564_24# VSS 0.95687f
.ends
```

LVS Check
On running following command
netgen -batch lvs "nand2b_layout.spice nand2b" "nand2b_sch.spice nand2b" /usr/local/share/pdk/sky130A/libs.tech/netgen/sky130A_setup.tcl

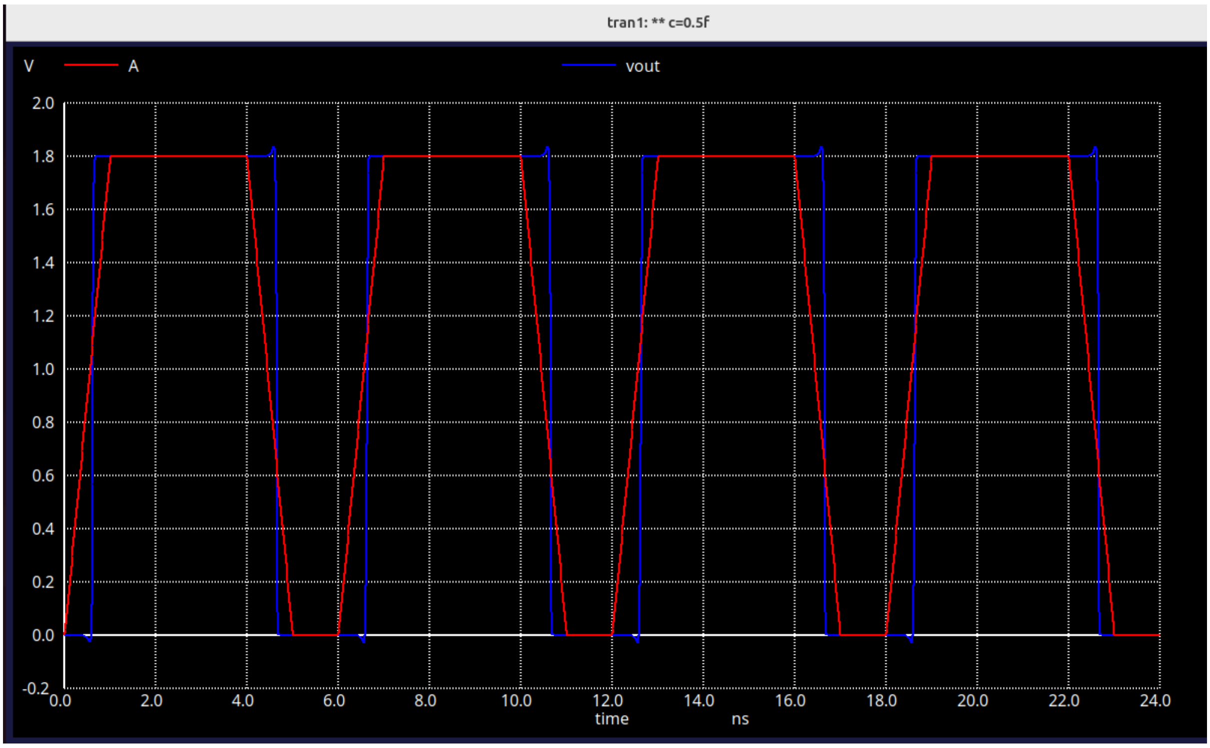
```
Circuit 1 contains 6 devices, Circuit 2 contains 6 devices.
Circuit 1 contains 7 nets,    Circuit 2 contains 7 nets.


Final result:
Circuits match uniquely.
.
Logging to file "comp.out" disabled
LVS Done.
```

DRC clean

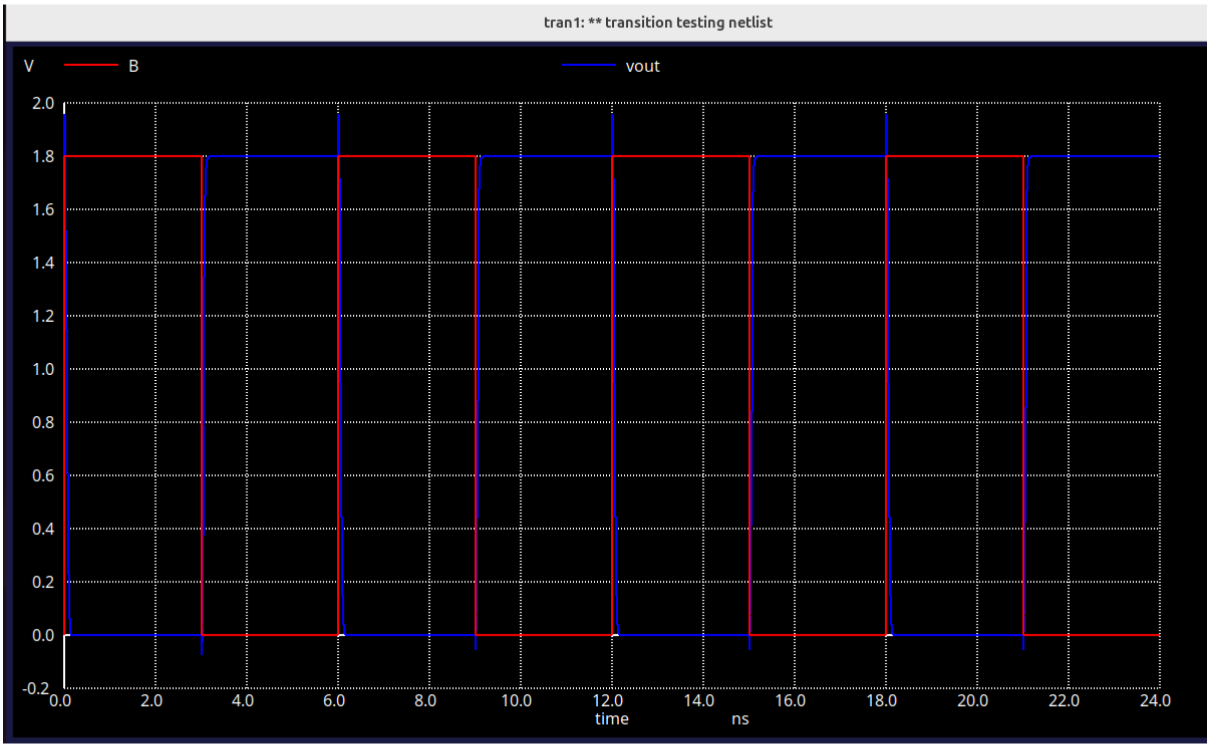




Input output waveforms which were used to rise, fall, and propagation delay.



C = 0.5fF, input pulse slew = 1000ps



C = 10fF input pulse slew = 10ps

Rise time was measured using following command

meas tran t20 when v(vout)=0.36 rise=2

meas tran t80 when v(vout)=1.44 rise=2

let tp = $&t80 - $&t20

Fall time was measured using following command

meas tran tf80 when v(vout)=1.44 fall=1

meas tran tf20 when v(vout)=0.36 fall=1

let tf = $&tf20 - $&tf80

Propagation delay

Rise

meas tran ti50 when A=0.9 rise=2

meas tran to50 when vout=0.9 rise=2

let tp = $&to50 - $&ti50

Fall

meas tran tfi50 when A=0.9 fall=1

meas tran tfo50 when vout=0.9 fall=1

let tpf = $&tfo50 - $&tfi50

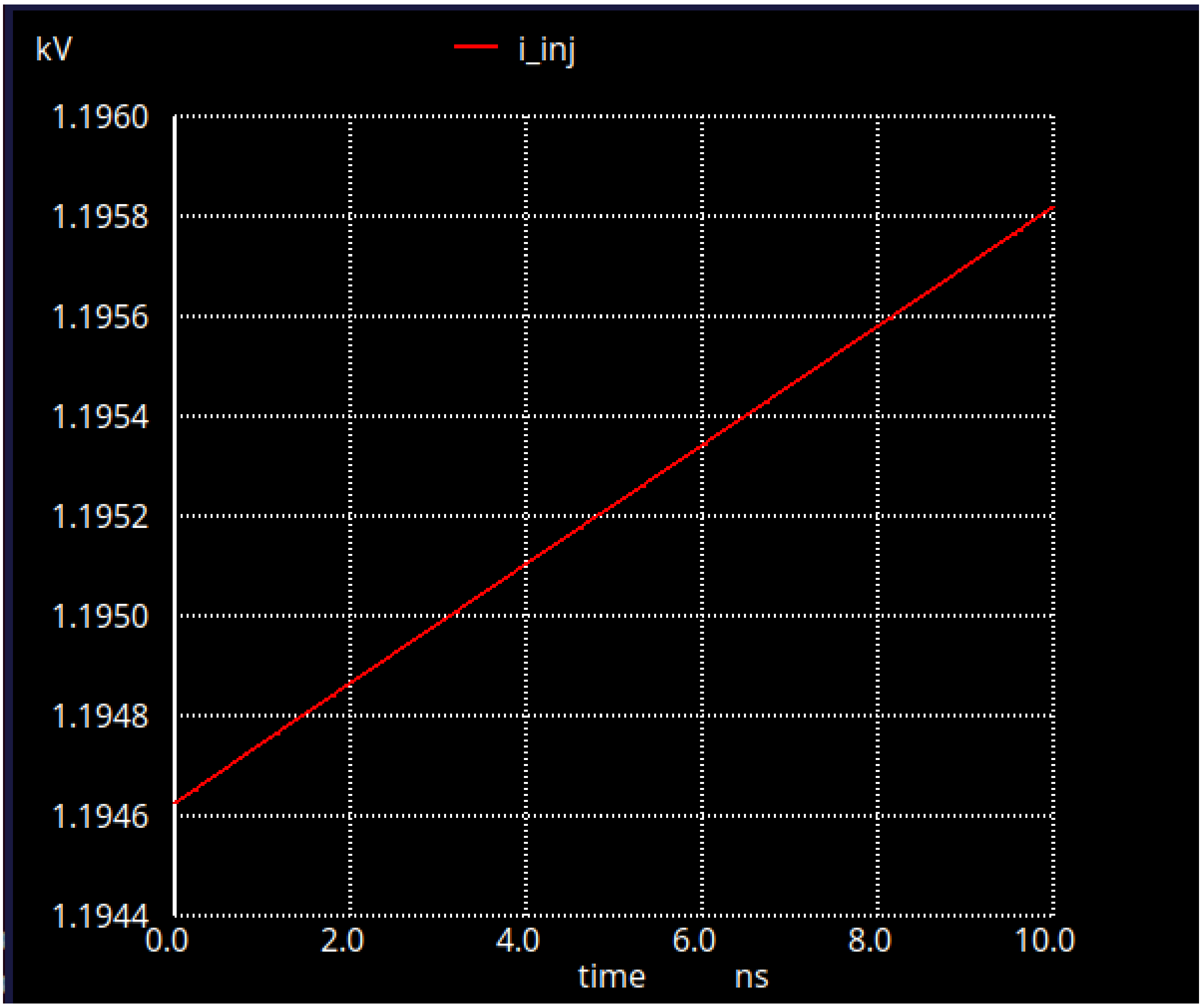
Input Capacitance

Input pin capacitances:

Input Pins	Rise Cap (pF)	Fall Cap (pF)	Average Cap (pF)
A	0.01073	0.01083	0.01078
B	0.00839	0.00845	0.00842

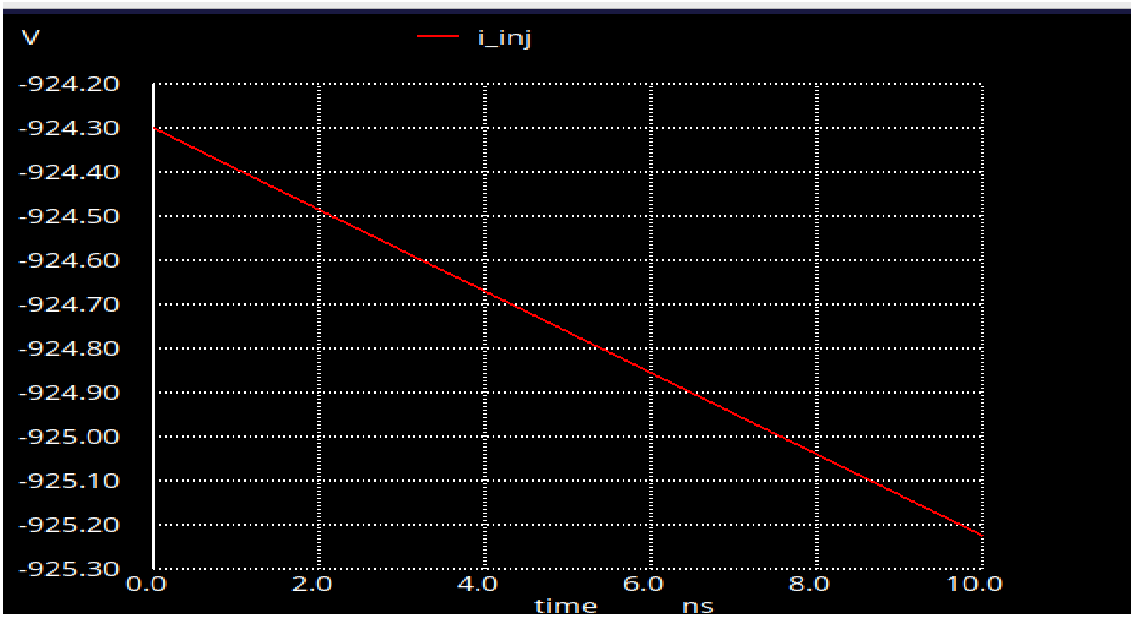
Input capacitance was measured by applying current source of 1uA at the Gate
Now we CV = It
Therefore, the slope of Voltage(V) vs time(t) graph gives I/C.
I is fixed at 1 uA.
Therefore C can be obtained as I/(Slope of V vs t graph)

The following V vs t graph is obtained when 1uA current is injected in Gate B of nand2b for measuring rise cap.



Slope = 1.192117e+08
Therefore, input capacitance of B gate = 1 uA / (Slope) = 0.00839 pF

Now for measuring fall capacitance
We with withdrawal current i.e negative current
And slope was obtained



The above graph was obtained for A gate
Slope = -9.23586e+07
C = I/slope = (-1 uA)/(Slope) = 0.01083 pF

Timing Tables

Transition Time Table: (considering 20% and 80% of VDD for transition time)

(i) Output Rise Transitions (in ps) [Input slew vs output capacitance].

Related pin A: (B at VDD)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.02094	0.01883	0.03477
10 fF	0.03612	0.03717	0.04912
100 fF	0.22777	0.22765	0.22870

Related pin B: (A at GND)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.02029	0.02809	0.10130
10 fF	0.03864	0.04276	0.14210
100 fF	0.21828	0.21828	0.30120

(ii) Output Fall Transitions (in ps) [Input slew vs output capacitance].

Related pin A: (B at VDD)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.02267	0.02175	0.03552
10 fF	0.04251	0.04381	0.05315
100 fF	0.26650	0.26640	0.26806

Related pin B: (A at GND)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.01821	0.02248	0.07928
10 fF	0.04319	0.04417	0.11218
100 fF	0.26647	0.26634	0.30660

Propagation delay time tables: (used 50% of input to 50% of output to simulate propagation delay – by keeping other inputs fixed).

Related pin A: (B at VDD)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.04151	0.05802	0.11516
10 fF	0.05613	0.07296	0.13614
100 fF	0.18818	0.20473	0.27070

Related pin B: (A at GND)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.01925	0.03972	0.11010
10 fF	0.03306	0.05474	0.15920
100 fF	0.15788	0.17933	0.39340

(ii) Cell Fall Delay (in ps) [Input slew vs output capacitance].

Related pin A: (B at VDD)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.04408	0.06324	0.14056

10 fF	0.06349	0.08293	0.16595
100 fF	0.23609	0.25549	0.34224

Related pin B: (A at GND)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	0.02856	0.04312	0.07971
10 fF	0.04699	0.06291	0.13671
100 fF	0.21889	0.23657	0.39665

Power Tables

Static Power (cover all possible input combinations based on number of inputs).

Condition (ABC)	**Power (nW)**
00	0.01236
01	1.76453
10	0.01236
11	1.76453

Dynamic Power Table:

(i) Rise Power (in nW) [Input slew vs output capacitance].

Related pin A: (B held constant at VDD)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	797.18	915.94	1063.46
10 fF	2354.95	2395.21	3001.17
100 fF	18408.1	18386.8	18388.2

Related pin B: (A held constant at 0)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	865.84	1277.3	1378.28
10 fF	2517.16	3109.42	3025
100 fF	18722.8	18720.5	19283.5

(ii) Fall Power (in nW) [Input slew vs output capacitance].

Related pin A: (B held constant at VDD)

	10 ps	**100 ps**	**1000 ps**
0.5 fF	828.08	883.78	1073.41
10 fF	828.58	840.64	1109.53
100 fF	864.94	842.24	1401.38

Related pin B: (A held constant at 0)

	10 ps	100 ps	1000 ps
0.5 fF	777.39	772.36	1018.86
10 fF	814.6	836.23	895.69
100 fF	851.82	841.93	760.89

Dynamic power calculation ngspice netlist screenshot

```
X1 A B VDD 0 Y nand2b
C1 Y 0 100f

Vin A 0 PULSE(0 1.8 0 1000p 1000p 3n 6n)
Vdd VDD 0 1.8
Vi_2 B 0 1.8

*Vin B 0 PULSE(0 1.8 0 1000p 1000p 3n 6n)
*Vdd VDD 0 1.8
*Vi_1 A 0 0

.tran 1ps 20ns 0 5ps
.control
run

meas tran tr20 when v(Y)=0.2*1.8 rise=1
meas tran tr80 when v(Y)=0.8*1.8 rise=1
meas tran curr_int_r integ vdd#branch from=tr20 to=tr80
let power_int_r=curr_int_r*1.8
print power_int_r/6e-09
meas tran tf80 when v(Y)=0.8*1.8 fall=1
meas tran tf20 when v(Y)=0.2*1.8 fall=1
meas tran curr_int_f integ vdd#branch from=tf80 to=tf20
let power_int_f=curr_int_f*1.8
print power_int_f/6e-09

.endc
.end
```

Static power calculation ngspice netlist screenshot

```
X1 A B VDD 0 Y nand2b
C1 Y 0 0.5f
Vdd VDD 0 1.8
Vi_1 A 0 0
Vi_2 B 0 0
.control
run
dc Vi_1 0 1.8 1.8
dc Vi_2 0 1.8 1.8
print vdd#branch
let power_static=vdd#branch*vdd
print power_static
.endc
.end
```

nand2b.v was compiled successfully

```
newuser@harsh:~/Project1/P1$ iverilog nand2b.v
newuser@harsh:~/Project1/P1$
```

Contribution Table

Name and Roll No.	Contribution
Harsh Gupta, 25M1237	nand2b design, timing and input capacitance characterization, buffer timing and input characterization, timing characterization of DFF, Verilog Codes and Report writing of nand2b.
Shubham, 25M1277	nand2b layout design, static and dynamic power characterization of nand2b and buffer cell and report writing of buffer cell
Ankit Raj 25M1269	Buffer design, Buffer layout,D flip flop design, Dflip flop layout, D report half
Ayush Singh25m1239	Input capacitance ,static and dynamic power for D flipflop.